University Management System

INTRODUCTION

The university needs a centralized system to manage:

- Students
- Professors
- Courses & Enrollment
- Departments
- Student Clubs
- Books & Borrowing
- Projects & Collaboration

BUSINESS REQUIREMENTS OVERVIEW

- Unique IDs for all main entities (Student, Professor, Course, Dept., Book, Club, Project)
- Students may enroll in multiple courses
- Professors belong to one department & may supervise or advise students
- Students may join clubs (optional)
- Students can borrow books from the library
- Projects require collaboration among students, professors, and departments
- Student has multi-valued attributes (phone numbers)

MAIN ENTITIES

- Student
- Professor
- Department
- Course
- Book & Library
- Club
- Project
- Academic Advisor

STUDENT ENTITY

Student Attributes:

- Student ID (PK)
- Student Name
- Email
- Date of Birth → used to calculate Age
- Phone Number (multi-valued)
- University ID Card (Card Number, Issue Date)

Relationships:

- Enrolls in Course (M:N)
- Borrows Books (1:M)
- Joins Clubs (M:N)
- Participates in Projects (via Collaboration)

PROFESSOR & DEPARTMENT

Professor Attributes:

- Prof_ID (PK)
- Name

Department Attributes

- Dept_ID (PK)
- Dept_Name

Relationships

- Professor assigned to Department (1:N)
- Department supervised by Professor (1:1 mandatory)
- Professor advises Student (1:N)
- Professor responsible for Course (1:N or N:M depending on your ERD)
- Professor supervises Projects

COURSES & ENROLLMENT

Course Attributes

- Course_ID
- Course Name

Department Attributes

- Dept_ID (PK)
- Dept_Name

Enrollment Relationship

- Student ↔ Course is Many-to-Many
- Relationship captures:
  - Grades
  - Semester (if planned)

Course Responsibility

- Professor ↔ Course

(your ERD shows “responsible” relationship, possibly M:N)

LIBRARY & BOOK BORROWING

Book Attributes

- Book_ID
- Book Title
- Author
- Library

Borrow Relationship

- Student borrows Book (1 Student → N Borrow records)

STUDENT CLUBS

Club Attributes

- Club_ID
- Club Name

Join Relationship

- Student ↔ Club is M:N
- Not mandatory for all students
- Clubs may have zero members

RESEARCH PROJECTS & COLLABORATION

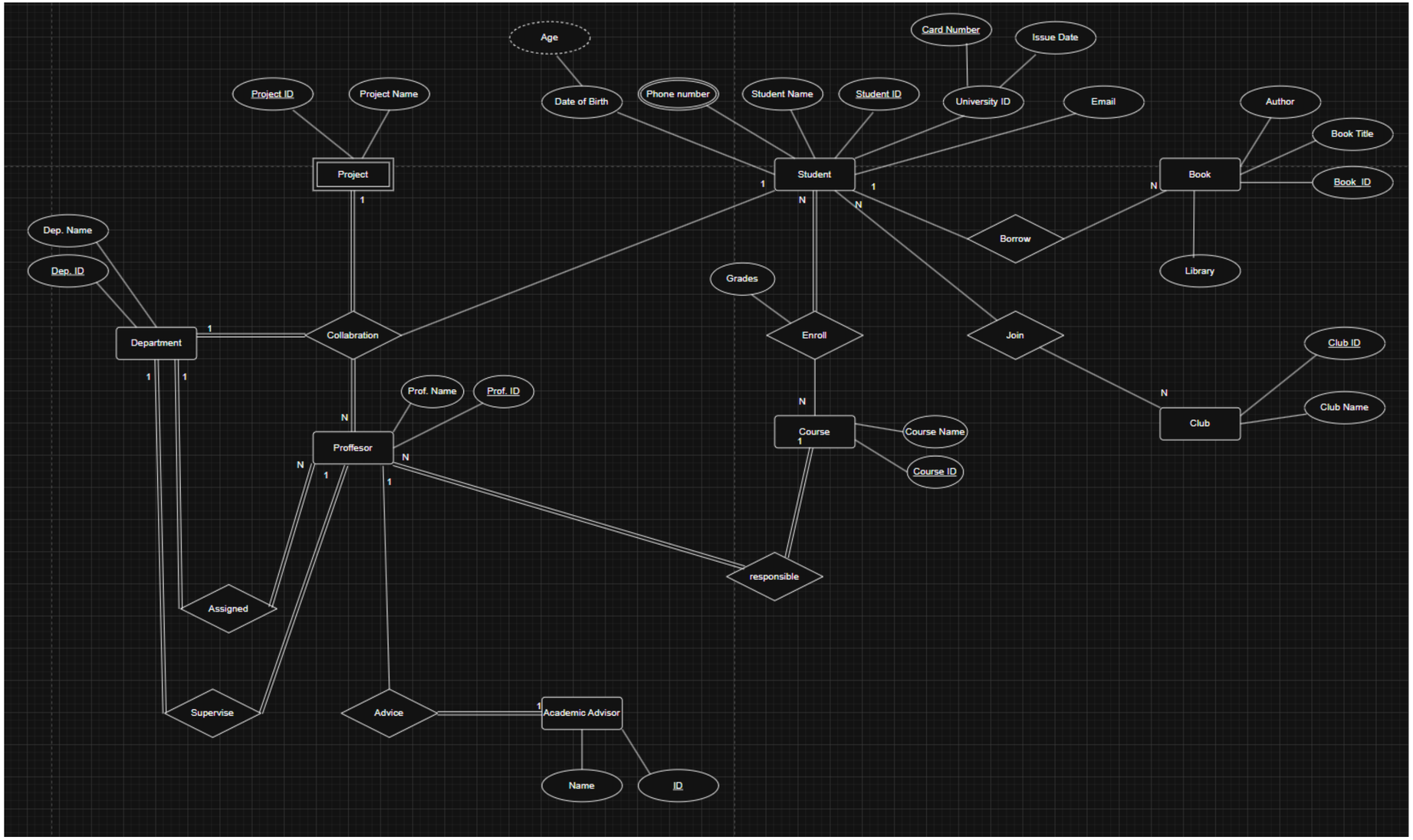
Project Attributes

- Project_ID
- Project Name

Collaboration Relationship Involves:

- Student
- Professor
- Department

FULL ERD DIAGRAM



Entity-Relationship Diagram (ERD) — Final Model

Thank You